MohamedAmine Azirgui

Rabat, Morocco
Email: mohamed.a.azirgui@gmail.com | Phone: +212 645336641
GitHub: github.com/maaaz5 | LinkedIn: linkedin.com/profile | Portfolio: mohamedamine.me

AI ENGINEER

AI Engineer focused on building intelligent applications and AI-powered systems. Experienced in NLP, embeddings, RAG, machine learning, and transformer-based architectures. Skilled in developing backend APIs and integrating AI models into production-ready applications, with a growing focus on agentic AI and LLM-based systems.

WORK EXPERIENCE

Software Engineer, Athing.online **Feb 2025 - Jan 2026**

- Built an end-to-end data pipeline using Python, Redis, Docker, and Supabase to ingest, clean, and process large-scale e-commerce product data
- Developed configurable web crawlers handling pagination, dynamic content, and unstructured HTML using CSS selectors and JSON-based configurations
- Implemented OpenAI-based extraction workflows for schema generation, data cleaning, and validation across heterogeneous product sources
- Designed embedding-based product matching system to detect duplicates and improve price comparison accuracy
- Deployed and maintained scalable crawling infrastructure on AWS with Docker and scheduled jobs, ensuring continuous data freshness

Full Stack Developer, Fikralabs **Mar 2022 - Sep 2023**

- Designed responsive dashboards and mobile UI/UX for web applications.
- Built dynamic charts, interactive components, and implemented consistent design systems.
- Integrated APIs and developed full-stack features using MERN stack and Next.js.

EDUCATION

Computer Engineering (Data Science specialization) **Oct 2023 - July 2026**

École Supérieure de Télécommunication, d'Informatique et de Management (SUPMTI)

Software Development Technician **Oct 2021 - July 2023**

Ecole Française d'Enseignement Technique (EFET)

PROJECTS

Bus Occupancy Level Prediction *PFE - Final Year Project*

- Built a physics-based simulation engine to generate realistic passenger occupancy data from raw GTFS transit schedules
- Merged raw static GTFS schedules with external contextual datasets (weather metrics, local event schedules, and 9 geographic stop profiles) to enrich features.
- Developed a dual-horizon ML system: XGBoost for long-term fleet planning and a 3-layer LSTM for real-time stop-level prediction
- Applied feature engineering including cyclic time encoding, target encoding for 363 stops, and relative route position
- Developed a full production stack: FastAPI backend, Next.js operator dashboard, and a React Native mobile app (iOS/Android)

GPT from Scratch – Text Classification (Spam Detection)

- Built a GPT-style transformer model from scratch using PyTorch
- Implemented attention mechanism, embeddings, and tokenization pipeline
- Trained model for spam classification (spam vs non-spam messages)
- Evaluated model performance and optimized training workflow

ADDITIONAL INFORMATION

Technical Skills

- Python, TypeScript, JavaScript, React, Next.js, NestJS, Machine Learning, Deep Learning, NLP, Transformers, Embeddings, Backend Development, REST APIs, GraphQL, Database Design (PostgreSQL, MongoDB), UI/UX, System Design Fundamentals, CI/CD, Git, RAG, Langchain, Agents, Langsmith

Languages

- English, German, French, Arabic.

Activities

- Selected to present web development projects during an official meeting with the German Federal Foreign Minister Annalena Baerbock in a formal institutional setting.